

Develop a program to manage resource allocation for five processes (Google Drive, Firefox, Word Processor, Excel, and PowerPoint) using four types of resources (Printer, ROM, Hard Disk, and RAM). The program takes input for allocated, maximum, and available resources, calculates the current need of each process, and determines if a safe execution order exists using the Banker's Algorithm.

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import numpy as np
import pandas as pd # Table sundar dikhane ke liye

processes = ['Google Drive', 'Firefox', 'Word', 'Excel', 'PPT']
resources = ['Printer', 'ROM', 'HDD', 'RAM']

# Inputs
allocation = np.array([[0, 1, 0, 0], [2, 0, 0, 1], [3, 0, 2, 2], [2, 1, 1, 0], [0, 0, 2, 2]])
max_demand = np.array([[7, 5, 3, 4], [3, 2, 2, 2], [9, 0, 2, 2], [2, 2, 2, 2], [4, 3, 3, 3]])
available = np.array([3, 3, 2, 2])

# 1. Need Matrix Calculate aur Print karna
need = max_demand - allocation
print("---- 1. NEED MATRIX (Max - Allocation) ----")
print(pd.DataFrame(need, index=processes, columns=resources))
print("\n" + "-"*40 + "\n")

# 2. Safety Algorithm Check
def check_safety():
    work = available.copy()
    finish = [False] * 5
    safe_sequence = []

    print("---- 2. STEP-BY-STEP EXECUTION ----")
    for _ in range(5):
        found = False
        for i in range(5):
            if not finish[i] and all(need[i] <= work):
                print(f"Executing {processes[i]}... (Available: {work} >= Need: {need[i]})")
                work += allocation[i]
                finish[i] = True
                safe_sequence.append(processes[i])
                print(f"Finished {processes[i]}. New Available: {work}\n")
                found = True
                break
        if not found: break

    if len(safe_sequence) == 5:
        print("---- 3. FINAL RESULT ----")
        print("✅ SYSTEM IS IN SAFE STATE!")
        print("SAFE SEQUENCE:", " -> ".join(safe_sequence))
    else:
        print("❌ SYSTEM IS IN UNSAFE STATE (DEADLOCK)!")

check_safety()

```

```

--- 1. NEED MATRIX (Max - Allocation) ---
      Printer  ROM  HDD  RAM
Google Drive    7    4    3    4
Firefox         1    2    2    1
Word            6    0    0    0
Excel           0    1    1    2
PPT             4    3    1    1

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--- 2. STEP-BY-STEP EXECUTION ---
Executing Firefox... (Available: [3 3 2 2] >= Need: [1 2 2 1])
Finished Firefox. New Available: [5 3 2 3]

Executing Excel... (Available: [5 3 2 3] >= Need: [0 1 1 2])
Finished Excel. New Available: [7 4 3 3]

Executing Word... (Available: [7 4 3 3] >= Need: [6 0 0 0])
Finished Word. New Available: [10 4 3 3]

Executing Google Drive... (Available: [10 4 3 3] >= Need: [7 4 3 4])
Finished Google Drive. New Available: [10 5 5 5]

Executing PPT... (Available: [10 5 5 5] >= Need: [4 3 1 1])
Finished PPT. New Available: [10 5 7 7]

--- 3. FINAL RESULT ---
✅ SYSTEM IS IN SAFE STATE!
SAFE SEQUENCE: Firefox -> Excel -> Word -> Google Drive -> PPT

```